

tional effort of the algorithm, regardless of how many measurements are available. Finally, Section 3.4 presents the Wiener filter, which is a method of estimating a time-varying signal that is corrupted by noise, on the basis of noisy measurements. Until 1960, Wiener filtering was the state of the art in signal estimation. The paradigm of signal estimation was shattered with the publication of Rudolph Kalman's work and related papers in the early 1960s, but it is still worthwhile understanding Wiener filtering because of its historical place in the history of signal estimation. Furthermore, Wiener filtering is still very useful in signal processing and communication theory.

3.1 ESTIMATION OF A CONSTANT

In this section, we will determine how to estimate a constant on the basis of several noisy measurements of that constant. For example, suppose we have a resistor but we do not know its resistance. We take several measurements of its resistance using a multimeter, but the measurements are noisy because we have a cheap multimeter. We want to estimate the resistance on the basis of our noisy measurements. In this case, we want to estimate a constant scalar but, in general, we may want to estimate a constant vector.

To put the problem in mathematical terms, suppose x is a constant but unknown n -element vector, and y is a k -element noisy measurement vector. How can we find the "best" estimate $\hat{x}$ of x ? Let us assume that each element of the measurement vector y is a linear combination of the elements of x , with the addition of some measurement noise:

$$\begin{aligned} y_1 &= H_{11}x_1 + \cdots + H_{1n}x_n + v_1 \\ &\vdots \\ y_k &= H_{k1}x_1 + \cdots + H_{kn}x_n + v_k \end{aligned} \quad (3.1)$$

This set of equations can be put into matrix form as

$$y = Hx + v \quad (3.2)$$

Now define ϵ_y as the difference between the noisy measurements and the vector $H\hat{x}$:

$$\epsilon_y = y - H\hat{x} \quad (3.3)$$

ϵ_y is called the measurement residual. As Karl Gauss wrote [Gau04], the most probable value of the vector x is the vector $\hat{x}$ that minimizes the sum of squares between the observed values y and the vector $H\hat{x}$. So we will try to compute the $\hat{x}$ that minimizes the cost function J , where J is given as

$$\begin{aligned} J &= \epsilon_{y1}^2 + \cdots + \epsilon_{yk}^2 \\ &= \epsilon_y^T \epsilon_y \end{aligned} \quad (3.4)$$

J is often referred to in control and estimation books and papers as a cost function, objective function, or return function. We can substitute for ϵ_y in the above equation to rewrite J as

$$\begin{aligned} J &= (y - H\hat{x})^T (y - H\hat{x}) \\ &= y^T y - \hat{x}^T H^T y - y^T H \hat{x} + \hat{x}^T H^T H \hat{x} \end{aligned} \quad (3.5)$$

In order to minimize J with respect to $\hat{x}$, we compute its partial derivative and set it equal to zero:

$$\begin{aligned}\frac{\partial J}{\partial \hat{x}} &= -y^T H - y^T H + 2\hat{x}^T H^T H \\ &= 0\end{aligned}\tag{3.6}$$

Solving this equation for $\hat{x}$ results in

$$\begin{aligned}H^T y &= H^T H \hat{x} \\ \hat{x} &= (H^T H)^{-1} H^T y \\ &= H^L y\end{aligned}\tag{3.7}$$

where H^L , the left pseudo inverse of H , exists if $k \geq n$ and H is full rank. This means that the number of measurements k is greater than the number of variables n that we are trying to estimate, and the measurements are linearly independent. In order to prove that we have found a minimum rather than some other type of stationary point² of J , we need to prove that the second derivative of J is positive semidefinite (see Problem 3.1).

■ EXAMPLE 3.1

Let us go back to our original problem of trying to estimate the resistance x of an unmarked resistor on the basis of k noisy measurements from a multimeter. In this case, x is a scalar so our k noisy measurements are given as

$$\begin{aligned}y_1 &= x + v_1 \\ &\vdots \\ y_k &= x + v_k\end{aligned}\tag{3.8}$$

These k equations can be combined into a single matrix equation as

$$\begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix} = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} x + \begin{bmatrix} v_1 \\ \vdots \\ v_k \end{bmatrix}\tag{3.9}$$

Equation (3.7) shows that the optimal estimate of the resistance x is given as

$$\begin{aligned}\hat{x} &= (H^T H)^{-1} H^T y \\ &= \left(\begin{bmatrix} 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_k \end{bmatrix} \\ &= \frac{1}{k} (y_1 + \cdots + y_k)\end{aligned}\tag{3.10}$$

In this simple example, we see that least squares estimation agrees with our intuition to simply compute the average of the measurements.

▽▽▽

²A stationary point of a function is any point at which its derivative is equal to zero. A stationary point of a scalar function could be a maximum, a minimum, or an inflection point. A stationary point of a vector function could be a maximum, a minimum, or a saddle point.